

AP Calculus AB, My Study Notes

Gerald Ellison. Mr. Osei, period 3. Practice exam Tuesday April 14, 2027. Real exam Monday May 3, 2027.

Limits and Continuity

- Plug in first. If you get a number, that is the limit.
- $0/0$ means factor, rationalize, or simplify, then plug in again.
- A function is continuous at a if the limit exists, $f(a)$ exists, and they are equal.
- Limits at infinity: compare the highest powers top and bottom.

Derivatives

- Power rule: $d/dx [x^n] = n \text{ times } x^{(n-1)}$.
- Product rule: $(fg)' = f'g + fg'$.
- Quotient rule: $(f/g)' = (f'g - fg') / g^2$. Low d high minus high d low, over low squared.
- Chain rule: $d/dx f(g(x)) = f'(g(x)) \text{ times } g'(x)$. This is the one I forget.
- Know cold: $d/dx \sin x = \cos x$, $d/dx \cos x = -\sin x$, $d/dx e^x = e^x$, $d/dx \ln x = 1/x$.

Applications of Derivatives

- Critical points where $f'(x) = 0$ or undefined.
- First derivative test for increasing or decreasing, second derivative test for concavity.
- Related rates: write the relationship, differentiate with respect to t , then plug in. Watch the units.
- Mean Value Theorem: if continuous and differentiable, there is a c where $f'(c)$ equals the average rate of change.

Integrals

- Antiderivative of x^n is $x^{(n+1)} / (n+1) + C$. Never forget the plus C .
- Fundamental Theorem: the integral from a to b of f equals $F(b)$ minus $F(a)$.
- U-substitution: pick u as the inside function, find du , swap, integrate.
- Area under the curve, average value of a function, and accumulation problems all use definite integrals.

Things I Keep Messing Up

- Forgetting the chain rule on the inside function.
- Dropping the plus C on indefinite integrals.
- Sign errors in the quotient rule.
- Reading related rates units wrong. Slow down and label everything.